

Franklin Wohnoutka

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EDUCATION

Drake University, Des Moines, IA

GPA: 3.82 | Dean's List | Expected Graduation: May 2027

Bachelor of Science

Majors: Data Analytics, Mathematics, and Computer Science | Minor: Physics

TECHNICAL SKILLS

Languages: Python, Java, SQL, R, C, JavaScript, NetLogo, SAS

Frameworks: PyTorch, TensorFlow (AlexNet, ResNet, VGGNet), React, React Native, Flask

Tools & Software: Git, Microsoft Office, MySQL, AWS (EC2, S3, RDS, Kinesis), Azure, Supabase, Expo, EAS

Concepts: Machine Learning, Database Design, Data Visualization, Multivariate Calculus, Cloud Systems

PROJECTS

Posted: Fits & Hauls – Fashion Social Media App (iOS) | App Store: [Posted-Link](#)

June 2026 – Present

- Independently designed and shipped a full-stack fashion sharing iOS app to the App Store (138 downloads), building a front end with React Native/Expo, a backend with Supabase/PostgreSQL, and native builds with EAS
- Designed a 16-table database with row-level security policies, real-time subscriptions, and PostgreSQL RPC functions for features such as friend discovery, follow requests, and streak resets
- Built core features including a social feed, wardrobe calendar, clothes tagging system, and theme customization

Random Search for Treasure on Connected Graphs – Mathematics Capstone

March 2026 – May 2026

- Formulated a random-search variant of the "tipsy cop and tipsy robber" graph game, in which a hunter searches a connected graph for buried treasure, and defined capture-time metrics to compare hunter starting positions
- Wrote a recursive Python program to compute expected capture times, observed an average of $(n - 1)/2$, and proved that for randomly placed treasure, capture time is uniform over 0 to $n - 1$ on connected n -vertex graphs

Immigration Enforcement Simulation using NetLogo

January 2026

- Built a NetLogo agent-based model (2,000 agents) simulating how immigration enforcement affects community fear and crime, calibrated to data from the February 2026 Minneapolis ICE raids; presented at DUCURS
- Analyzed 10 trials per setting across 100-4,000 officers, finding enforcement did not reduce crime but increased community fear and the fear gap between targeted and non-targeted groups

RELEVANT EXPERIENCE

Research Assistant

M.I.S.S.F.I.T (Magneto-Ionization Spacecraft Shield for Interplanetary Travel), Des Moines, IA

May 2025 – Present

- Lead the computational group of a student-led research team designing a magnetic and ionization radiation shield for crewed interplanetary travel, coordinating the work of 4 students; presented at DUCURS
- Built a relativistic Monte Carlo tracker simulating cosmic ray trajectories through a magnetic field, with SRIM-derived energy loss and particle data gathered from SOHO/ACE repositories
- Created software that solves the Laplace equation, employing an iterative relaxation method to generate the 3-D B-field, and ran simulations of varying field strength to determine the minimum effective strength

Tutor

Drake University Mathematics & Computer Science Tutoring Lab, Des Moines, IA

November 2025 – Present

- Tutor students from precalculus through Calculus IV and introductory programming through Machine Learning, using guided problem solving and worked examples rather than direct answers
- Brief professors on student progress and flag topics needing additional support

EXTRACURRICULAR ACTIVITIES

Secretary, Society of Physics Students, Des Moines, IA

February 2025 – Present

Member, Wallin Scholars (scholarship, advising, and career program), Minneapolis, MN

August 2023 – Present